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NUMERICAL METHODS

CPE 312

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Lecturer



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MULLER'S METHOD



Learning Objectives

- *Define Muller's Method
- *Determine and perform the algorithm of Muller's Method
- *Solve functions using Muller's Method



- **Müller's Method** is an iterative numerical technique used to find roots of a function, typically of the form $f(x)=0$.
- It is an extension of the secant method, employing a quadratic interpolation to estimate roots, which allows for faster convergence, especially when dealing with complex or non-linear functions.



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- Unlike the secant method, which relies on a linear approximation between two points, Müller's method uses three points and fits a quadratic polynomial to them.
- It can converge to complex roots even if the initial guesses are real, making it versatile for both real and complex functions.



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- It requires three initial guesses, x_0 , x_1 , and x_2 , which are used to calculate subsequent approximations to the root.
- It is particularly useful in engineering and applied mathematics, where functions can have multiple roots, including complex ones, and a high rate of convergence is desired.



Muller's Method Algorithm

1. Select three initial guesses x_0 , x_1 , and x_2 near the expected root.
2. Compute $f(x_0)$, $f(x_1)$, and $f(x_2)$.
3. Calculate divided differences.



Distance between the chosen points

$$h_1 = x_1 - x_0 \qquad h_2 = x_2 - x_1$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1}$$

$$\delta_2 = \frac{f(x_2) - f(x_1)}{h_2}$$



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$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1}$$

$$b = a \cdot h_2 + \delta_2$$

$$c = f(x_2)$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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$$\varepsilon_{a^3} = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$

Relative percent error
Stopping criterion (0.0...)



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Example-1

Find the root of $f(x) = x^3 - x - 1$
using Muller's Method.



Step 1.

$$f(x) = x^3 - x - 1$$

$$f(0) = 0^3 - 0 - 1 = -1$$

$$f(1) = 1^3 - 1 - 1 = -1$$

$$f(2) = 2^3 - 2 - 1 = 5$$

Initial Interval **[1, 2]**



Step 2.

$$X_2 = \frac{a+b}{2}$$

$$X_2 = \frac{1+2}{2}$$

$$X_2 = \frac{3}{2}$$

$$\mathbf{x_2 = 1.5}$$

$$f(1.5) = (1.5)^3 - 1.5 - 1 = \mathbf{0.875}$$

[illegible]



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875					

$$h_1 = x_1 - x_0$$



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875					

$$h_1 = x_1 - x_0$$

$$h_1 = 2 - 1 = 1$$



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875					

$$h_1 = x_1 - x_0$$

$$h_1 = 2 - 1 = 1$$

$$h_2 = x_2 - x_1$$



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875					

$$h_1 = x_1 - x_0$$

$$h_1 = 2 - 1 = 1$$

$$h_2 = x_2 - x_1$$

$$h_2 = 1.5 - 2 = -0.5$$



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875					

$$h_1 = x_1 - x_0$$

$$h_1 = 2 - 1 = 1$$

$$h_2 = x_2 - x_1$$

$$h_2 = 1.5 - 2 = -0.5$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1}$$



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875					

$$h_1 = x_1 - x_0$$

$$h_1 = 2 - 1 = 1$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1} = 6$$

$$h_2 = x_2 - x_1$$

$$h_2 = 1.5 - 2 = -0.5$$



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
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$$h_1 = x_1 - x_0$$

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1	1	2	1.5	-1	5	0.875					

$$h_1 = x_1 - x_0$$

$$h_1 = 2 - 1 = 1$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1} = 6$$

$$h_2 = x_2 - x_1$$

$$h_2 = 1.5 - 2 = -0.5$$

$$\delta_2 = \frac{f(x_2) - f(x_1)}{h_2} = 8.25$$



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875					

$$h_1 = 1$$

$$h_2 = -0.5$$

$$\delta_1 = 6$$

$$\delta_2 = 8.25$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1}$$

$$b = a \cdot h_2 + \delta_2$$

$$c = f(x_2)$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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1	1	2	1.5	-1	5	0.875					

$$h_1 = 1$$

$$h_2 = -0.5$$

$$\delta_1 = 6$$

$$\delta_2 = 8.25$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.5$$

$$b = a \cdot h_2 + \delta_2$$

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$$c = f(x_2) = 0.875$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.5$$

$$b = a \cdot h_2 + \delta_2 = 6$$

$$c = f(x_2) = 0.875$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}} = 1.33333$$



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$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}} = 1.33333$$



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1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	

$$\varepsilon_a^3 = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$



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n	x_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ϵ_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	

$$\epsilon_a^3 = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$

$$\epsilon_a^3 = 12.50028$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = x_1 - x_0$$

$$h_2 = x_2 - x_1$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = x_1 - x_0 = -0.5$$

$$h_2 = x_2 - x_1$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = x_1 - x_0 = -0.5$$

$$h_2 = x_2 - x_1 = -0.16667$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = x_1 - x_0 = -0.5$$

$$h_2 = x_2 - x_1 = -0.16667$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1}$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = x_1 - x_0 = -0.5$$

$$h_2 = x_2 - x_1 = -0.16667$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1} = 8.25$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = x_1 - x_0 = -0.5$$

$$h_2 = x_2 - x_1 = -0.16667$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1} = 8.25 \quad \delta_2 = \frac{f(x_2) - f(x_1)}{h_2}$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = x_1 - x_0 = -0.5$$

$$h_2 = x_2 - x_1 = -0.16667$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1} = 8.25$$

$$\delta_2 = \frac{f(x_2) - f(x_1)}{h_2} = 5.02778$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = -0.5$$

$$h_2 = -0.16667$$

$$\delta_1 = 8.25$$

$$\delta_2 = 5.02778$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ϵ_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = -0.5$$

$$h_2 = -0.16667$$

$$\delta_1 = 8.25$$

$$\delta_2 = 5.02778$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1}$$

$$b = a \cdot h_2 + \delta_2$$

$$c = f(x_2)$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702					

$$h_1 = -0.5$$

$$h_2 = -0.166667$$

$$\delta_1 = 8.25$$

$$\delta_2 = 5.02778$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.833331$$

$$b = a \cdot h_2 + \delta_2$$

$$c = f(x_2)$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331				

$$h_1 = -0.5$$

$$h_2 = -0.16667$$

$$\delta_1 = 8.25$$

$$\delta_2 = 5.02778$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.83331$$

$$b = a \cdot h_2 + \delta_2$$

$$c = f(x_2)$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331				

$$h_1 = -0.5$$

$$h_2 = -0.166667$$

$$\delta_1 = 8.25$$

$$\delta_2 = 5.02778$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.83331$$

$$b = a \cdot h_2 + \delta_2 = 4.22221$$

$$c = f(x_2)$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ϵ_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221			

$$h_1 = -0.5$$

$$h_2 = -0.16667$$

$$\delta_1 = 8.25$$

$$\delta_2 = 5.02778$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.83331$$

$$b = a \cdot h_2 + \delta_2 = 4.22221$$

$$c = f(x_2)$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221			

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$$b = a \cdot h_2 + \delta_2 = 4.22221$$

$$c = f(x_2) = 0.03702$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ϵ_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702		

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$$\delta_2 = 5.02778$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.83331$$

$$b = a \cdot h_2 + \delta_2 = 4.22221$$

$$c = f(x_2) = 0.03702$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}} = 1.32447$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	

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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	

$$\varepsilon_a^3 = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	

$$\varepsilon_a^3 = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$

$$\varepsilon_a^3 = 0.66895$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	0.66895

$$\varepsilon_a^3 = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$

$$\varepsilon_a^3 = 0.66895$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	0.66895
3	1.5	1.33333	1.32447								



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	0.66895
3	1.5	1.33333	1.32447	0.875							



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	0.66895
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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	0.66895
3	1.5	1.33333	1.32447	0.875	0.03702	-0.00106					



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ϵ_a^3
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	0.66895
3	1.5	1.33333	1.32447	0.875	0.03702	-0.00106					

$$h_1 = x_1 - x_0 = -0.16667$$

$$h_2 = x_2 - x_1 = -0.00886$$

$$\delta_1 = \frac{f(x_1) - f(x_0)}{h_1} = 5.02778 \quad \delta_2 = \frac{f(x_2) - f(x_1)}{h_2} = 4.29797$$



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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ε_{a^3}
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	0.66895
3	1.5	1.33333	1.32447	0.875	0.03702	-0.00106					

$$h_1 = -0.16667 \quad a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.15775$$

$$h_2 = -0.00886$$

$$b = a \cdot h_2 + \delta_2 = 4.26113$$

$$\delta_1 = 5.02778 \quad c = f(x_2) = -0.00106$$

$$\delta_2 = 4.29797 \quad x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}} = 1.32472$$

$$\varepsilon_{a^3} = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$

$$\varepsilon_{a^3} = 0.01887$$



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1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
2	2	1.5	1.33333	5	0.875	0.03702	4.83331	4.22221	0.03702	1.32447	0.66895
3	1.5	1.33333	1.32447	0.875	0.03702	-0.00106	4.15775	4.26113	-0.00106	1.32472	0.01887

$$h_1 = -0.16667 \quad a = \frac{\delta_2 - \delta_1}{h_2 + h_1} = 4.15775$$

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$$b = a \cdot h_2 + \delta_2 = 4.26113$$

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$$\delta_2 = 4.29797 \quad x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}} = 1.32472$$

$$\varepsilon_{a^3} = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$
$$\varepsilon_{a^3} = 0.01887$$



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1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
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n	X_0	x_1	x_2	$f(x_0)$	$f(x_1)$	$f(x_2)$	a	b	c	x_3	ϵ_{a^3}
1	1	2	1.5	-1	5	0.875	4.5	6	0.875	1.33333	12.5
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3	1.5	1.33333	1.32447	0.875	0.03702	-0.00106	4.15775	4.26113	-0.00106	1.32472	0.01887

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$$c = f(x_2) = -0.00106$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}} = 1.32472$$

$$\epsilon_{a^3} = \left| \frac{x_3 - x_2}{x_3} \right| \times 100\%$$
$$\epsilon_{a^3} = 0.01887$$



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The approximate root of
 $f(x) = x^3 - x - 1$ is 1.32472.



ACTIVITY # 9

1. Use the Muller's Method to find the root of $f(x) = 2x^3 - 2x - 5$.



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